

Gitesh Chawda

Machine Learning Engineer

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SUMMARY

Machine Learning Engineer with production software experience and deep open-source work across the transformer, large language model, and computer vision ecosystem. Sole maintainer of KerasFormers, a multi-backend Keras 3 library porting 60+ vision, language, and vision-language architectures with on-the-fly Hugging Face weight conversion, and author of 42 merged contributions to core deep learning libraries including Hugging Face Transformers and Keras.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, SQL

ML and Deep Learning: PyTorch, TensorFlow, Keras 3, JAX, Hugging Face Transformers, scikit-learn, Transformers, Flash Attention, Diffusion Models, Computer Vision

LLMs, Inference, and Fine-tuning: vLLM, TensorRT, Triton, CUDA, XLA, bitsandbytes, Quantization, LoRA, PEFT, SFT, GRPO, KV Cache Optimization, Model Serving

Agentic AI Frameworks: LangChain, LangGraph, LlamaIndex, CrewAI, AutoGen, OpenAI Agents SDK, MCP, A2A

MLOps and Cloud: Docker, Linux, Git, GitHub Actions, CI/CD, MLOps, LLMops, AWS, GCP

Web and Data: React.js, Flask, GraphQL, Node.js, NumPy, Pandas, OpenCV, MySQL, MongoDB, Oracle

PROJECTS AND OPEN SOURCE

KerasFormers, Open-source, pip-installable Keras 3 library for Vision, LLM, VLM, Depth, and Speech models

[Github](#)

- Built KerasFormers, a backend-agnostic Keras 3 library supporting 60+ architectures across vision, speech, LLM, and VLM tasks (SAM, DETR, DINOv3, Whisper, Llama, Qwen, DeepSeek, Mistral, Gemma, GLM), with single-codebase execution on TensorFlow, PyTorch, and JAX.
- Engineered Hugging Face-to-Keras checkpoint conversion with bit-close parity validation against reference implementations, and developed Keras 3 native weight-only quantization (INT8, INT4, FP8) using backend-agnostic operations, enabling up to 8x model size reduction for efficient inference deployment.

Open-Source Contributions, 42 merged pull requests across Hugging Face Transformers, Keras, Keras.io, and KerasCV

- Contributed 42 merged pull requests across Hugging Face Transformers, Keras, Keras.io, and KerasCV, improving model utilities, tokenizer modularity, type safety, and developer workflows.
- Added core Keras 3 capabilities including new multi-backend ops (PSNR, unravel_index, diagflat, inner, exp2), augmentation layers, YOLOv8 CIoU loss/backbone tests, and official keras.io examples.

txtcode, Open-source TypeScript npm package for controlling AI coding agents remotely from any messaging app [Github](#)

- Built a tool that orchestrates agentic coding assistants (Claude Code, OpenAI Codex, Cursor, Gemini CLI, Ollama, and more) from 6 messaging platforms (WhatsApp, Telegram, Discord, Slack, Teams, Signal), dispatching natural-language coding tasks to an agent on a remote machine with no port forwarding or VPN.
- Designed a provider-agnostic abstraction over 10 LLM backends (Anthropic, OpenAI, Gemini, Mistral, xAI Grok, and others), with mid-session hot-swapping of providers and coding adapters via slash commands and per-session logging for live monitoring.

EXPERIENCE

Fiserv, Software Developer, Chennai, India (July 2023 - August 2024)

- Developed and maintained a production web application with React.js, GraphQL, and AWS (AppSync, Lambda, API Gateway, Cognito, S3), supporting internal banking and payments workflows.
- Reduced unauthorized access attempts by 85% by strengthening authentication with AWS AppSync and Amazon Cognito, improving overall security posture.
- Secured fintech infrastructure by remediating 25+ critical vulnerabilities surfaced through internal audits and penetration testing, alongside the application security team.

Braynix AI, Deep Learning Intern, Bangalore, India (September 2021 - December 2021)

- Built deep learning pipelines for medical X-ray abnormality detection, improving validation accuracy by 20% over baseline CNN models across 5,000+ images.
- Reached 92% precision and 0.85+ IoU in medical image segmentation by implementing a U-Net with an EfficientNet-B4 encoder.
- Trained and benchmarked 3 object detection models (YOLOv5, SSD, RetinaNet) on 1,000+ labeled images at 0.78+ mAP.

EDUCATION

California State University, Fullerton, Master of Science in Computer Science, California, USA (August 2024 - May 2026)

Pimpri Chinchwad College of Engineering, Bachelor of Engineering in Computer Engineering, Pune, India (August 2019 - July 2023)

PUBLICATIONS

EffiConvRes: An Efficient Convolutional Neural Network with Residual Connections and Depthwise

Convolutions, 2023 7th International Conference on Computing, Communication, Control and Automation (ICCUBEA).

Survey and Comparison of Various Pre-trained CNN Architectures and CNN-Transformer Combinations, Smart Grid and Electric Vehicle Conference.